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PFAS levels in paired drinking water and serum samples collected from an exposed community in Central North Carolina

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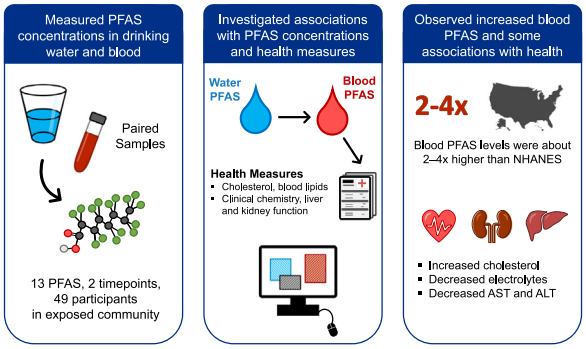
HIGHLIGHTS

- PFAS were measured in the blood and drinking water of an exposed community in NC.
- Some blood PFAS concentrations were 2–4 times higher than the general U.S. population.
- Serum PFHxA increased between timepoints, reflecting temporal changes in water concentrations.
- Serum PFOA and PFHxS concentrations were positively associated with cholesterol.
- Serum PFHxA was negatively associated with liver enzymes (AST and ALT).

GRAPHICAL ABSTRACT



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ABSTRACT

The community of Pittsboro, North Carolina has been documented to have extensive per- and polyfluoroalkyl substances (PFAS) contamination in its drinking water source, the Haw River, over the last 20 years. However, a detailed exposure assessment has never been conducted. In this study, we sought to characterize the PFAS in paired drinking water and blood samples collected from a small cohort of Pittsboro residents ($n = 49$). Drinking water and serum from blood were collected from adults in late 2019 and early 2020 and were analyzed to quantify 13 PFAS analytes. In order to explore potential health effects of PFAS exposure, serum was further analyzed for clinical chemistry endpoints that could be potentially associated with PFAS (e.g., cholesterol, liver function biomarkers). PFAS were detected in all serum samples, and some serum PFAS concentrations were 2 to 4 times higher than the median U.S. serum concentrations reported in the general U.S. population. Of the 13 PFAS in drinking water, perfluorohexanoic acid (PFHxA) was measured at the highest concentrations. PFAS levels in the current drinking water were not associated with current serum PFAS, suggesting that the serum PFAS in this cohort likely reflects historical exposure to PFAS with long half-lives (e.g., PFOS and PFOA). However, one PFAS with a shorter half-life (PFHxA) was observed to increase in serum, reflecting the temporal variability of PFHxA in river and drinking water. Statistical analyses indicated that serum PFOA and PFHxS were positively associated with total cholesterol and non-HDL cholesterol. No serum PFAS was associated with HDL cholesterol. In the clinical chemistry analyses, serum PFHxA was found to be negatively associated with electrolytes and liver enzymes (e.g., AST and ALT), and serum PFOS was found to be positively

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associated with the ratio of blood urea nitrogen to creatinine (BUN:Cre). While small in size, this study revealed extensive exposure to PFAS in Pittsboro and associations with clinical blood markers, suggesting potential health impacts in community residents.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are widespread environmental contaminants linked to adverse health effects in humans (ATSDR, 2021). PFAS are especially notable as recalcitrant contaminants in drinking water systems across the United States (Andrews and Naidenko, 2020; Hu et al., 2016; Quinones and Snyder, 2009). Due to their unique chemical properties and environmental persistence, PFAS are difficult to remove from water with standard drinking water treatment and filtration (Herkert et al., 2020). These PFAS in drinking water are a significant contributor to human PFAS exposure (Hoffman et al., 2011; Hu et al., 2019; Post et al., 2012; Sunderland et al., 2019).

Much research has focused on the health effects of a specific legacy PFAS called PFOA or C8; PFOA has been well-established as a drinking water contaminant of health concern. In Parkersburg, West Virginia, drinking water was so highly contaminated with PFOA from an industrial facility that the serum PFOA levels of some nearby residents were up to 500% higher than the general U.S. population (Frisbee et al., 2009; Frisbee et al., 2010). The subsequent epidemiologic investigation, the C8 Health Project, recruited over 69,000 participants in West Virginia and Ohio who had been drinking this PFOA-contaminated water; this study observed associations between PFAS exposure and adverse health outcomes (Steenland et al., 2020), such as kidney and testicular cancer (Barry et al., 2013), ulcerative colitis (Steenland et al., 2013), increased serum lipids (Frisbee et al., 2010; Steenland et al., 2009), altered liver function biomarkers (Gallo et al., 2012), and changes in thyroid hormones (Knox et al., 2011). Other communities have also been affected by PFAS contamination in their drinking water (Banzhaf et al., 2017; Landsteiner et al., 2014; Li et al., 2018; Lindstrom et al., 2011; Xu et al., 2020; Zhang et al., 2016).

Previous research has heavily concentrated on the legacy chemicals PFOA and PFOS because of their toxicity and their persistence in both the environment and in people. However, there is a dearth of information on the newer, emerging PFAS that are now replacing PFOA and PFOS. While the shorter-chain perfluoroalkyl acids (PFAAs) such as PFPeA, PFHxA, and PFHpA are thought to be less persistent in the human body, they are poorly characterized in terms of their known exposure, potential toxicity, or environmental persistence. This data gap is a pertinent issue for people in North Carolina who are currently being exposed to these short-chain PFAAs through their drinking water (Herkert et al., 2020).

Surface water across North Carolina has been impacted by PFAS contamination, particularly in the Cape Fear River watershed. In 2017, an emerging PFAS known as GenX (or hexafluoropropylene oxide dimer acid, HFPO-DA) was frequently detected in tap water in Wilmington, North Carolina. The source of Wilmington's tap water – the Cape Fear River – was found to be contaminated with a variety of legacy and emerging PFAS (Sun et al., 2016a). The Haw River, a tributary of the Cape Fear River, is also impacted by many of the same PFAS found in the Cape Fear River, including short-chain perfluoroalkyl acids (PFAAs) (Pétré et al., 2022). The Haw River serves as the source of drinking water to the community of Pittsboro, a small community in central North Carolina with a population of approximately 4,000 residents. As conventional water treatment does not remove PFAS, much of the PFAS in the Haw River can also be found in Pittsboro's finished drinking water (Herkert et al., 2020). There is also concern that the historical water exposure to harmful legacy PFAS in these communities may be substantial. In 2006, the Haw River had water concentrations of PFOS at 132 ng/L and of PFOA at 287 ng/L (Nakayama et al., 2007).

In addition to the high levels of PFOA and PFOS, other studies have detected high levels of PFPeA and PFHpA in the Cape Fear River watershed as early as 2012 (Strynar et al., 2015). Research from our group has noted that the water concentrations of these shorter-chain PFAAs are temporally variable (Pétré et al., 2022). This temporal variability in PFAS concentrations in Haw River water (and subsequently in Pittsboro tap water) is driven by seasonal changes in river discharge and flow, seasonal changes in evaporation and precipitation, and temporal and sporadic changes in effluent PFAS concentrations or industrial activities (Pétré et al., 2022). Variable levels of PFAS in drinking water may have implications for PFAS exposure in the Pittsboro community. This is especially true for the shorter-chain PFAAs which have human elimination half-lives on the order of days and months. This short timeframe may be limiting our ability to capture the full health consequences of long-term exposure to these shorter-chain PFAS. Additionally, these specific PFAS in drinking water have demonstrated some adverse health effects in an animal model; pregnant rabbits showed adverse effects on the liver, kidney, and placenta after exposure to drinking water with a PFAS profile identical to Pittsboro tap water (Crute et al., 2022).

The Pittsboro community is thus unique because its residents are currently being exposed to temporally-variable levels of newer, short-chain PFAAs through drinking water. The goal of this study was to evaluate the concentrations of PFAS in paired samples of both drinking water and serum of Pittsboro residents over time. By using paired samples collected at two different timepoints in late 2019 and early 2020, we aimed to understand how the contribution of changing drinking water PFAS concentrations would affect blood PFAS concentrations, particularly for short-chain PFAAs. Additionally, a further goal was to evaluate whether blood PFAS was associated with any changes in health measures, such as increased serum cholesterol.

2. Materials and methods

2.1. Study population

Between fall 2019 and early 2020, participants were recruited into our cross-sectional study to evaluate the levels of PFAS in the drinking water and serum of Pittsboro, North Carolina residents. The Duke University Institutional Review Board reviewed and approved all study protocols prior to recruitment. Eligible participants all met the following inclusion criteria: over 18 years of age and living in Pittsboro, North Carolina at the time of enrollment. Exclusion criteria included those who could not safely give blood.

A total of 49 participants provided informed consent and were enrolled into the study. Participants were asked to complete a questionnaire asking about their age, sex, menstrual status, water source, water consumption, water filtration, and number of years they had lived in their current home. The study consisted of two timepoints, approximately 2–3 months apart (November 2019 and January/February 2020); at each timepoint participants provided paired samples of serum and drinking water.

2.2. Sample collection, extraction, purification, and LC-MS/MS analysis

2.2.1. Drinking water sample collection and PFAS analysis

Participants provided a sample of their primary household drinking water (e.g., tap water, filtered water, bottled water) at each timepoint. They were supplied pre-cleaned, labeled, wide-mouth, one-liter high-density polyethylene (HDPE) bottles (Catalog No: 414004-115, Lot No. VHF120219, VWR International). Before use, sample bottles were pre-cleaned with rinses of methanol and ammonium hydroxide, methanol alone, and ultra-pure water. Participants were instructed to uncap the bottle, rinse the bottle three times with their drinking water, fill the bottle to ~0.5 inches below the bottom rim (~1.0 L), and return the water bottle

immediately to the study team. Water samples were stored on wet ice or reusable ice packs until extraction and LC-MS/MS analysis.

Analysis of water for 13 PFAS was performed as described in Herkert et al. (2020). In brief, for each water sample, a volume of 800 mL was spiked with isotopically-labeled internal standards (Wellington Laboratories, Guelph, Canada) and processed through an automated solid-phase extraction (SPE) system (Dionex AutoTrace 280 Solid-Phase Extraction instrument; Sunnyvale, California, USA). A list of the 13 PFAS measured is in Supplemental Table S1, and further methods information on the water extraction and analysis can be found in Supplemental Material.

2.2.2. Serum sample collection and PFAS analysis

Participants provided a non-fasting blood sample (approximately 10 mL) which was collected by a certified phlebotomist into a serum-separator tube (BD Vacutainer® SST™ tube, 10.0 mL, Becton-Dickinson (BD), Franklin Lakes, New Jersey). Whole blood was allowed to clot over wet ice. Blood tubes were stored on wet ice for no more than three hours before centrifugation to separate blood into serum. Centrifugation was performed in a swinging bucket centrifuge (Eppendorf™ 5810R with Rotor A-4-62, rotor radius of 18.0 cm) at room temperature for 5 min at 3500 rpm (2465 rcf). Serum was immediately aliquoted into polypropylene cryo-vials (Nalgene®, ThermoScientific, Catalog No: 5000-0020 and 5000-1020) and stored at −80 °C until analysis. Further methods information on the serum extraction and analysis can be found in Supplemental Material.

2.3. Quality assurance and quality control (QA/QC)

For both serum and water samples, laboratory processing blanks of LC-MS grade water were included in each batch and extracted alongside samples. For quality assurance and control with the serum measurements, a serum standard reference material (SRM) from the National Institute of Standards and Technology (NIST) was used. Aliquots of approximately 1.0 mL of reconstituted SRM 1958 Fortified Freeze-Dried Human Serum (NIST, 2018) were analyzed alongside samples. PFAS concentrations measured in SRM 1958 are reported in Supplemental Table S4, and the observed serum SRM measurements have strong concordance with NIST reference values; PFNA in SRM was measured at 82–132% of the NIST value and all other PFAS were between 88–119% of the NIST values.

Isotopically-labeled internal standards (^{13}C- and ^{18}O-mass labeled, listed in Supplemental Table S2) were used in all samples. A mass-labeled recovery standard (M2-PFOA; also $^{13}C_2$-PFOA) was added to all samples immediately before LC-MS/MS analysis. Recovery of mass-labeled PFAS was calculated to assess the recovery efficiency of the extraction and clean-up methods; recovery for all analytes in serum and water at each timepoint are reported in Supplemental Table S5.

2.4. Serum lipid panel and comprehensive metabolic panel measurements

Serum from blood was analyzed by a clinical laboratory, LabCorp (Burlington, North Carolina), to obtain information on potential clinical health measures. Serum samples were kept frozen at −80 °C until they were submitted to the Burlington, North Carolina LabCorp facility for analysis in October 2020. Serum was analyzed for a panel of lipids. Total cholesterol, triglycerides, and high-density lipoprotein (HDL) cholesterol were measured, and low-density lipoprotein (LDL) and very-low-density (VLDL) lipoprotein cholesterol were calculated as part of the serum lipid panel. A comprehensive metabolic panel (CMP) was also performed on the serum, also by LabCorp. Further methods information on the serum lipid panel and comprehensive metabolic panel measurements can be found in Supplemental Material.

2.5. Statistical analysis

Serum and water PFAS concentration values were blank-subtracted using the average of all the laboratory processing blank values, and method detection limits (MDLs) were calculated as three times the standard deviation of the laboratory processing blank values for each analyte. Analytes detected in ≤50% of water or blood samples at either timepoint were excluded from further statistical analysis. For analytes with a detection frequency >50%, values less than MDL were imputed with MDL divided by two.

We examined the distributions of all data. Because both serum PFAS concentrations and water PFAS concentrations were lognormally distributed, we conducted non-parametric analyses (e.g., Mann-Whitney *U* test, Spearman correlation) or $\log_{10}$-transformed these data before inclusion in models as described below. Some serum clinical measures were also lognormally distributed and similarly $\log_{10}$-transformed prior to analysis.

We evaluated how blood PFAS concentrations differed between variables such as sex and age by first averaging each participants' serum concentration values from both timepoints, if data were available. We then performed Mann-Whitney *U* tests to explore if serum concentration PFAS differed based on sex, menstrual status, age, or years living in the current home. We dichotomized age into two categories at the median age of 58 years (58 or younger vs. 59 and older), and we dichotomized years in home into two categories (7 or more years vs. <7 years).

Because samples were collected at two timepoints, further statistical analyses were performed as either repeated-measures in linear mixed effects models or stratified by timepoint. A total of 92 samples were collected. Timepoint 1 had a sample size of $n = 48$ participants while Timepoint 2 had $n = 44$. A total of 43 participants attended both timepoints and provided blood and water at the two timepoints.

We evaluated the correlations between water PFAS and serum PFAS at different times using Spearman correlations with all data stratified by timepoint. In order to further explore the associations between water PFAS and serum PFAS, we conducted multivariate repeated-measures analyses using the water and serum concentrations for individual PFAS analytes. These analyses included other covariates such as age, sex, and years in current home. Serum PFAS and water PFAS concentrations were always evaluated as individual analytes in separate models (e.g., multiple PFAS chemicals in serum were never included together in the same model).

We also examined relationships between serum PFAS analytes and age, serum lipids, and serum clinical measures by conducting Spearman correlations (stratified by timepoint). For serum lipid analysis, statistical analyses focused on total, HDL, and non-HDL cholesterol as these measures were independent of participants' fed state. Triglyceride measurements may be affected by fed state and since our participants were non-fasting, the triglyceride data were dropped from later regression analysis. LDL and VLDL cholesterol were also dropped from the later regression analysis because they were not measured directly and were calculated using participants' triglyceride measurements. For serum clinical chemistry measures, all 18 measured and calculated parameters were evaluated.

In order to more rigorously explore these associations between serum PFAS and health measure outcomes while accounting for possible confounding factors (e.g., sex), we performed repeated-measures mixed model analysis. Models for serum health measure outcomes were adjusted for age as a continuous variable and sex as a categorical variable (male vs. female). Models for serum PFAS concentration outcomes additionally included years in current home (<7 or 7+ years) as a covariate of interest. Measures that were lognormally distributed were $\log_{10}$-transformed before analysis (i.e., A:G ratio, albumin, ALT, AST, bilirubin, calcium, creatinine, and glucose). Serum PFAS concentrations were always evaluated as individual analytes in separate models (e.g., multiple PFAS chemicals in serum were never included together in the same model). A conceptual model is included in Supplemental Fig. S1 to depict the predictors, outcomes, and included covariates. Statistical analyses were performed using JMP Pro 16.0.0, GraphPad Prism 9.3.1, and Microsoft Excel. Mixed models were performed with JMP Pro.

2.6. Sensitivity analyses

The effect of menstrual status (eumenorrhea vs. amenorrhea) on model results was evaluated as a categorical variable (three categories: male vs. non-menstruating female vs. menstruating female); a sensitivity analysis

was performed and showed no qualitative changes in model performance (data not shown). Menstrual status did not qualitatively affect our results or conclusions and given the low sample size (i.e., $n = 6$ participants with normal eumenorrhea), this variable was dropped from further analysis in mixed models.

Data were $\log_{10}$-transformed to improve normality where applicable and to reduce the potential influence of outliers. Outlier testing was performed with all measurements included in mixed models (ROUT method, Q = 1%) (Motulsky and Brown, 2006). No outliers were identified for any $\log_{10}$-transformed serum PFAS concentrations or $\log_{10}$-transformed water PFAS concentrations. However, a few outliers were identified for some serum clinical endpoints, even after $\log_{10}$-transformation. We conducted sensitivity analyses excluding these potential outliers, and results and conclusions were unaffected (data not shown). Therefore, all results presented here include all measured values.

For the analysis of water PFAS concentrations, values less than MDL were imputed with MDL divided by two. Because the MDLs differed between the two timepoints, we conducted a sensitivity analysis to determine if the imputation impacted our conclusions about water concentration changes over time. We conducted sensitivity analyses for the analytes with significant changes over time (PFBA, PFPeA, PFHxA, and PFHpA; Fig. S3) and applied the higher MDL to all 92 samples across both timepoints and imputed a uniform value (data not shown). This sensitivity analysis confirmed the same conclusion that the water concentration for those analytes did indeed decrease between timepoints and was not simply an artifact of the higher MDL at timepoint 2.

3. Results and discussion

3.1. Study population

The study cohort was primarily female and older, with a median age of 58 years. Complete demographic data are included in Supplemental Table S6 and visually depicted in Supplemental Fig. S2.

3.2. Drinking water PFAS concentrations

Water PFAS concentration data are reported in Table 1. We observed some changes over time in drinking water PFAS concentrations. Water concentrations for several short-chain perfluorocarboxylic acids (PFCAs) decreased between T1 and T2 as shown in Table 1 and Supplemental Fig. S3. Four analytes (PFBA, PFPeA, PFHxA, and PFHpA) were significantly higher at the first timepoint (T1), than at the second timepoint (T2). A large proportion of samples at T2 were less than the MDL for most PFAS analytes.

The decline in PFCAs in drinking water between the timepoints (over three months) may be driven by the decline in the source water's PFAS concentrations; PFAA concentrations in the Haw River declined between November 2019 and January 2020 (Pétré et al., 2022) and Haw River PFAS concentrations are typically reflective of subsequent Pittsboro tap water PFAS. Alternatively, we cannot rule out the possibility that some of the decline in water PFAS concentration seen at Timepoint 2 may also be due to changes in water filtration (i.e., more participants installing water filters, upgrading to better water filters, and switching to bottled water, as seen in Table S6).

The drinking water samples provided by participants were generally higher in short-chain PFCAs than PFOA or PFOS (median levels of 12–34 ng/L vs. 2–5 ng/L at the first timepoint). These shorter-chain PFAS do not currently have any U.S. EPA drinking water health advisories. In contrast, PFOA and PFOS do have proposed regulatory limits and advisories. Proposed maximum contaminant levels (MCLs) issued by EPA in 2023 aim to establish an enforceable drinking water limit of 4.0 ng/L for PFOA and 4.0 ng/L for PFOS (USEPA, 2023). While MCLs are enforceable regulatory standards, other non-enforceable public health goals based solely on health effects have also been issued for PFOA and PFOS. These goal levels include the 2022 interim drinking water health advisories previously issued by EPA in June 2022, which were set at 0.004 ng/L for PFOA and 0.02 ng/L for PFOS (USEPA, 2022), and the 2023 proposed maximum contaminant level goals (MCLGs) which essentially updates both these

Table 1
Drinking water PFAS concentration data. Detection frequency, method detection limit (MDL), median, and 90th percentile concentrations of PFAS in drinking water samples at both timepoints (T1 and T2). The range for Σ(13)PFAS for all 92 observations was <MDL to 458 ng/L. The nine analytes highlighted were used in later analyses because they were detected in >50% of samples at both timepoints.

Analyte	Timepoint 1 (T1) (n=48)				Timepoint 2 (T2) (n=44)				Averaged (n=49)	
	Detection Frequency	MDL (ng/L)	Median (ng/L)	90th Percentile (ng/L)	Detection Frequency	MDL (ng/L)	Median (ng/L)	90th Percentile (ng/L)	Median (ng/L)	90th Percentile (ng/L)
PFBA	85%	0.11	14.3	50.7	68%	0.17	1.8	5.0	8.7	28.0
PFPeA	81%	0.18	28.5	98.5	64%	0.29	10.2	20.7	20.3	64.3
PFHxA	81%	0.16	34.3	161.1	66%	0.18	5.9	19.7	23.3	95.9
PFHpA	81%	0.09	12.6	72.7	61%	0.18	3.3	19.8	9.4	49.8
PFOA	90%	0.02	5.0	8.7	55%	0.38	0.7	6.8	2.8	7.9
PFNA	77%	0.04	0.5	0.7	46%	0.09	<MDL	0.7	—	—
PFDA	65%	0.09	0.3	0.5	36%	0.12	<MDL	0.2	—	—
PFBS	81%	0.29	4.1	6.0	66%	0.12	2.2	7.3	3.0	6.4
PFHxS	79%	0.04	1.7	3.1	57%	0.07	0.3	2.5	1.2	2.7
PFOS	75%	0.14	2.4	4.0	39%	0.72	<MDL	4.3	—	—
GenX	67%	0.01	0.04	0.09	80%	0.01	0.02	0.03	0.03	0.05
4:2 FTS	48%	0.002	<MDL	0.02	43%	0.01	<MDL	0.01	—	—
6:2 FTS	77%	0.04	0.7	1.2	57%	0.02	0.06	0.6	0.4	0.8

<MDL: less than method detection limit.
For analytes detected in both timepoints, participants' averaged values over both timepoints are reported; before averaging, values less than MDL were imputed as MDL divided by two.

health-based values to zero, indicating that there is no safe level of PFOS or PFOA in drinking water.

Using the 2023 proposed MCLs (i.e., the enforceable regulatory standards), we observed in our data that the median levels of PFOA in Pittsboro drinking water exceeded the proposed PFOA MCL limit of 4.0 ng/L (Table 1). The median levels of PFOA detected in our participants' drinking water were 5 ng/L and 0.7 ng/L at the two timepoints, indicating that PFOA in drinking water ranged from 175 to 1250 times higher than the June 2022 EPA health advisory. Median levels of PFOS in drinking water were 2.4 ng/L and non-detectable (Table 1) at the two timepoints, respectively (which is 120 times higher than the June 2022 EPA health advisory for PFOS). These PFOA and PFOS drinking water concentrations are of concern for human health because they exceed the interim health advisories by several orders of magnitude. The temporal variability seen from these two timepoints may indicate additional cause for concern; infrequent water sampling for PFAS may not be sufficient in detecting PFAS contamination. Pittsboro has previously seen excessive discharges and sporadic spikes of another industrial contaminant, 1,4-dioxane, in the Haw River (McElroy et al., 2019; Sun et al., 2016b). Frequent testing of water may be advisable to ensure that communities do not miss spikes in PFAS drinking water contamination due to increased industrial pollution or low river discharge volume (Pétré et al., 2022).

3.3. Serum PFAS concentrations in residents

Serum PFAS concentration data are reported in Table 2. Of the 13 analytes measured in serum, 6 PFAS were detected in 100% of participants at both timepoints (PFHxA, PFOA, PFNA, PFDA, PFHxS, and PFOS). An additional PFAS analyte, PFHpA, was detected at a lower frequency (i.e., 44% and 61% at the two timepoints; Table 2); remaining analytes were detected rarely in <5% of serum samples.

Median serum PFHxA concentrations increased by approximately 2.6 times from the first timepoint to the second timepoint as shown in Table 2 and Supplemental Fig. S4. All other serum PFAS concentrations were stable between timepoints and did not significantly increase or decrease. This difference among analytes in serum is likely because of the differences in elimination half-lives between different PFAS. Long-chain PFAS have serum elimination half-lives of years in humans and are thus less likely to decrease appreciably over the course of 2–3 months (Chiu et al., 2022; Fenton et al., 2021; Olsen et al., 2007). In contrast, the short-chain PFHxA has an estimated human serum half-life of approximately 32 days (between 14 and 49 days) (Dawson et al., 2023; Fenton et al., 2021; Russell et al., 2013). The short half-life of PFHxA, coupled with the temporally variable concentrations of PFHxA in drinking water (Table 1; Supplemental Fig. S3) and in the Haw River, including a high spike in river PFHxA concentration in early December 2019 (Pétré et al., 2022), may explain the increase in serum PFHxA seen in Table 2 and Supplemental Fig. S4.

As seen in the supplemental material of Pétré et al. (2022), the Haw River experienced a spike in concentration for several short-chain PFAS (PFHxA, PFHpA, PFPeA) around the week of 2 December 2019. The concentration of PFHxA in the river at that time exceeded 300 ng/L. These river water samples were collected downstream of the Burlington wastewater treatment plant at a location upstream of Pittsboro's drinking water intake. Our hypothesis is that our sample collection times in November 2019 and January/February 2020 were not frequent enough to detect the spike in drinking water PFAS pollution; thus, we don't see increases in drinking water concentrations. Infrequent but high spikes in chemical pollution have been previously observed in the Haw River for chemicals like 1,4-dioxane (McElroy et al., 2019; Sun et al., 2016b). However, because the serum half-life for PFHxA is relatively short (~32 days), we were able to detect an increase in median serum PFHxA (0.5 to 1.3 ng/g) when we sampled in January/February 2020.

Since PFHpA also spiked in the Haw River in December 2019 (~100 ng/L) (Pétré et al., 2022), we hypothesized that PFHpA might increase in serum like PFHxA. We observed an increase in the detection frequency of serum PFHpA (44 to 61%), but due to the high MDL for serum PFHpA at Timepoint 1 and the very low concentrations of serum PFHpA in general, we cannot say that serum PFHpA changed appreciably in our study population or that it changed in response to the levels of PFHpA in either river or drinking water.

The concentrations of PFAS measured in our participants' serum are several times higher than those reported in the general U.S. population. Over

Table 2
Blood serum PFAS concentration data. Detection frequency, method detection limit (MDL), median, and 90th percentile concentrations of PFAS in serum at both timepoints (T1 and T2). The six analytes highlighted were used in later analyses because they were detected in >50% of samples at both timepoints.

Analyte	Timepoint 1 (T1) (n=48)				Timepoint 2 (T2) (n=44)				Averaged (n=49)	
	Detection Frequency	MDL (ng/g)	Median (ng/g)	90th Percentile (ng/g)	Detection Frequency	MDL (ng/g)	Median (ng/g)	90th Percentile (ng/g)	Median (ng/g)	90th Percentile (ng/g)
PFBA	4%	0.56	<MDL	0.5	0%	0.67	ND	ND	—	—
PFHxA	**100%**	**0.45**	**0.5**	**0.8**	**100%**	**0.36**	**1.3**	**2.7**	**0.94**	**1.8**
PFHpA	44%	0.43	<MDL	1.1	61%	0.09	0.2	0.9	0.24	0.8
PFOA	**100%**	**0.06**	**6.4**	**15.1**	**100%**	**0.51**	**5.3**	**11.6**	**6.1**	**16.1**
PFNA	**100%**	**0.14**	**1.3**	**2.9**	**100%**	**0.33**	**1.5**	**2.6**	**1.4**	**3.5**
PFDA	**100%**	**0.25**	**0.7**	**1.2**	**100%**	**0.29**	**0.7**	**1.1**	**0.7**	**1.2**
PFBS	0%	0.5	ND	ND	0%	0.5	ND	ND	—	—
PFHxS	**100%**	**0.02**	**2.9**	**5.6**	**100%**	**0.1**	**2.6**	**5.2**	**2.96**	**6.0**
PFOS	**100%**	**0.13**	**10.7**	**21.3**	**100%**	**0.15**	**9.8**	**21.3**	**11.6**	**21.6**
GenX	0%	0.5	ND	ND	0%	0.35	ND	ND	—	—
4:2 FTS	0%	0.06	ND	ND	0%	0.02	ND	ND	—	—
6:2 FTS	0%	0.5	ND	ND	0%	0.01	ND	ND	—	—

PFPeA was not able to be reliably quantified in serum and is not reported; ND: not detected; <MDL: less than method detection limit.
For analytes detected in both timepoints, participants' averaged values over both timepoints are reported; before averaging for PFHpA, values less than MDL were imputed as MDL divided by two.

95% of the general U.S. population has PFAS in their blood as measured by the National Health and Nutrition Examination Survey (NHANES) (Calafat et al., 2019; Calafat et al., 2007; Calafat et al., 2007b; Ye et al., 2018). The median PFAS serum concentrations in our cohort are 2–4 times higher than the median levels for the general U.S. population as reported by NHANES and as shown in Fig. 1. The NHANES data plotted in Fig. 1 are from U.S. adults during survey years 2017–2018, and the blood from Wilmington residents was collected from adults in 2017–2018 (Kotlarz et al., 2020). The serum PFAS concentrations in our cohort are comparable to levels found in people living downstream in Wilmington, North Carolina. Both Wilmington and Pittsboro source their drinking water from surface water in the Cape Fear River Basin.

Previous literature has suggested that menstruation could be an elimination route for PFAS (Barton et al., 2020; Harada et al., 2005; Lorber et al., 2015; Singer et al., 2018; Wong et al., 2014) and that serum PFAS is lower in females and people who menstruate (Ericson et al., 2007; Kato et al., 2011). We evaluated blood PFAS concentrations stratified by sex and menstrual status. When stratifying the data by sex alone, females had statistically lower levels of blood PFHxA than males (Supplemental Fig. S5), but there was no difference for other PFAS between males and females. Further stratifying the data by menstrual status indicated that menstruating females had significantly lower levels of blood PFOS than males (Supplemental Fig. S6); however, our sample size for menstruating females was only $n = 6$, limiting our ability to detect differences.

Blood PFAS concentrations were also evaluated by age and by the number of years a participant lived in their current home. We hypothesized that older participants and participants who had lived in their home longer would have experienced PFAS exposure for a longer time and would thus have higher serum PFAS. We found that older participants had significantly higher serum PFOA and PFNA than younger participants (Supplemental Fig. S7). We also found that participants who had lived in their homes for 7 or more years had significantly higher serum PFOA, PFNA, and PFDA (Supplemental Fig. S8). These results suggest that residents who lived in Pittsboro for longer periods of time were exposed to more PFAS through their drinking water.

3.4. Water PFAS and serum PFAS comparisons and associations

To evaluate the differences in the relative composition of PFAS between blood and water, we compared the profiles of PFAS in each matrix in Supplemental Fig. S9. Despite the observation that PFOA and PFOS levels were relatively low in the drinking water samples (<5% of the total PFAS), PFOA and PFOS were the most abundant PFAS in the serum (~75% of the total PFAS). Of the 9 PFAS frequently detected in Pittsboro drinking water, only 3 (PFOA, PFHxA, and PFHxS) were detected in >50% of the paired serum samples.

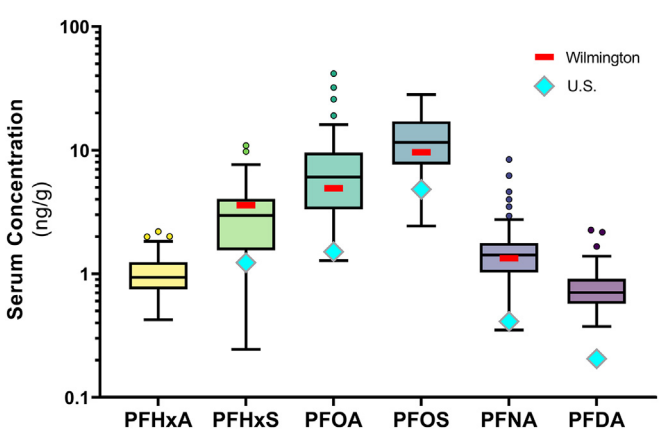

Fig. 1. Averaged serum PFAS concentrations in Pittsboro residents (n = 49). Data are compared with serum concentrations in the U.S. general population and residents of Wilmington, North Carolina. Data are represented in Tukey boxplots. The U.S. data (blue diamonds) are the median concentrations found in adults 20 years of age or older as reported in the CDC's NHANES survey years 2017–2018. The Wilmington data (thick red bars) are the median concentrations found in adults as reported in Kotlarz et al. (2020). In Kotlarz et al. (2020), PFDA was not measured, and PFHxA was detected in <60% of samples and is not reported. Concentrations are reported on logarithmic scale.

One goal of this study was to evaluate the relationship between drinking water PFAS exposure and corresponding serum PFAS. Spearman correlations calculated for water PFAS levels and serum PFAS levels indicated a relatively weak (though not statistically significant) association between water and serum concentrations at either timepoint (Supplemental Fig. S10). While each individual water PFAS was well-correlated with the other water PFAS at the same timepoint, they were less correlated with the serum PFAS concentrations (Supplemental Fig. S10). Most of the serum PFAS detected in this study were well-correlated with the other serum PFAS; however, PFHxA was unique and was notably less correlated with any of the other five detected serum PFAS (Supplemental Fig. S10). There was also some indication in our data that serum PFHxS was negatively correlated with the water PFAS concentrations at the first timepoint, but these were weak correlations, were not statistically significant, and were not repeated at the second timepoint (Supplemental Fig. S10).

We additionally conducted repeated-measures analyses to determine if drinking water PFAS concentrations were associated with corresponding blood PFAS levels after adjusting for age (continuous), sex (male vs. female), and years in current home (<7 or 7+ years). PFAS concentrations in water were not significantly associated with serum PFAS concentration in any of our models (Table 3). Blood PFAS was dominated by the more persistent PFAS with longer half-lives such as PFOA, possibly from previous drinking water exposure and not current water exposure. Bolstering this hypothesis, participants who had lived in their homes for 7 or more years were significantly more likely to have higher blood PFOA than participants who had lived <7 years in their home ($p < 0.05$; Supplemental Fig. S8). Additionally, years in home was a significant covariate in the model for PFOA (Table 3). Previous studies have detected high levels of PFOA in Pittsboro's drinking water source, the Haw River, as far back as 2006 (Nakayama et al., 2007; Sun et al., 2016a).

It is worth noting previous studies in the literature that have explored the relationship between drinking water PFAS and blood PFAS concentrations. While there are remaining data gaps for shorter-chain PFAS, the relationships for blood and water PFOA and PFOS have been extensively studied (Bartell et al., 2010; Gribble et al., 2015; Hoffman et al., 2011; Shin et al., 2011). Bartell (2017) describes how serum PFOA can be accurately estimated by using tap water PFOA concentrations in pharmacokinetic modeling, and such models can now be applied to predict serum PFOA, PFOS, PFNA, and PFHxS. These models are conveniently accessible in simple online calculators (https://www.ics.uci.edu/~sbartell/pfascalc.html; https://www.atsdr.cdc.gov/pfas/bloodlevelestimator/index.html).

3.5. Serum lipids

Correlations between blood PFAS and serum lipids were evaluated using Spearman correlations (Supplemental Fig. S11) with data stratified by timepoint (T1, $n = 48$; T2, $n = 44$; total, $n = 92$). Age was positively associated with HDL cholesterol at T1 ($p = 0.02$, $R_s = 0.33$) and PFOA, PFOS, PFNA, and PFDA at both T1 and T2 (all $p < 0.05$). PFOA at T2 was positively associated with total cholesterol ($p = 0.047$, $R_s = 0.30$). PFHxS at T2 was positively associated with non-HDL cholesterol ($p = 0.04$, $R_s = 0.31$). PFHxA at T2 was negatively associated with HDL cholesterol ($p = 0.01$, $R_s = -0.38$). Additionally, PFHxA at T1 and T2, and PFHxS, PFOA, and PFNA at T2 were positively associated with triglycerides ($p < 0.05$).

Study participants were not instructed to fast before blood was collected. Because triglyceride levels are sensitive to whether a person is fasting, analysis of triglyceride data in this study may be limited. This issue also impacts the LDL and VLDL cholesterol data as these are calculated using the triglyceride values; for example, VLDL cholesterol is simply estimated as 20% of the triglyceride value by the Friedewald formula. Previous

Table 3
Regression analysis of water PFAS predictors of PFAS in serum. Models were adjusted for age, sex, and years in home. The beta-coefficients (β), 95% confidence intervals, and adjusted *p*-values are reported. Both serum and water PFAS were $\log_{10}$-transformed before analysis. Only PFHxA, PFHxS, and PFOA are included because these were the only analytes detected in >50% of both blood and water samples.

	Serum PFAS					
	PFHxA		PFHxS		PFOA	
Predictor	β	p	β	p	β	p
Respective PFAS water concentration[1,2]	−0.03 (−0.07, 0.02)	0.21	0.02 (−0.005, 0.05)	0.10	0.01 (−0.01, 0.03)	0.49
Age[3,4]	0.2 (−0.5, 1.0)	0.51	0.9 (−0.5, 2.3)	0.19	**1.5 (0.2, 2.8)**	***0.03**
Sex						
Males	Reference	–	Reference	–	Reference	–
Females[4]	**−13.6 (−23.2, −2.8)**	***0.02**	−18.0 (−33.7, 1.5)	0.07	−14.2 (−30.0, 5.2)	0.14
Years in home						
7+ years	Reference	–	Reference	–	Reference	–
<7 years[4]	−1.8 (−12.2, 9.7)	0.74	−17.1 (−32.4, 1.7)	0.07	**−28.5 (−41.2, −13.0)**	***0.001**

[1] Each analyte was compared to itself (i.e., water PFHxA was used in models for serum PFHxA).
[2] Beta-coefficients represent the % change in serum PFAS for each 1% increase in water PFAS.
[3] Age is defined as a continuous variable (in years).
[4] Exponentiated beta-coefficients represent the % change in serum PFAS relative to the reference group for categorical variables, or per 1-year increase for age.
* $p < 0.05$.

human PFAS studies have found that fasting participants had markedly lower triglycerides than non-fasting participants (Steenland et al., 2009).

However, total cholesterol and HDL cholesterol are independent of fasting or fed state. The HDL cholesterol is referred to as "good" cholesterol for its beneficial effect on atherosclerosis. Non-HDL cholesterol (the subtraction of HDL cholesterol from total cholesterol) is a representative measure of the negative, potentially atherogenic effects of higher levels of cholesterol while accounting for the beneficial effect of HDL cholesterol (Frost and Havel, 1998) and is a better predictor of coronary heart disease risk than LDL cholesterol (Liu et al., 2006). Non-HDL cholesterol is also more robust than estimated LDL cholesterol as a health measure because it does not require any assumptions regarding VLDL cholesterol or triglycerides values (which are impacted by fasting) (Frost and Havel, 1998).

The associations between serum PFAS and serum lipids were further evaluated through repeated-measures mixed models; analyses were adjusted for age and sex. Results indicated that total cholesterol and non-HDL cholesterol were positively associated with both PFHxS and PFOA as shown in Fig. 2 and Supplemental Table S7. These results suggest that for every 10% increase in serum PFHxS, there is, on average, a 2 mg/dL increase in total cholesterol. HDL cholesterol was not associated with any individual serum PFAS (Fig. 2; Table S7).

Our results are similar to other studies that have found associations between PFOA and PFOS with increased total and non-HDL cholesterol and have found no association between PFOA and PFOS with HDL cholesterol (Dong et al., 2019; Nelson et al., 2010; Steenland et al., 2020; Steenland et al., 2009). Associations between serum PFAS and cholesterol have also been observed in a similarly-exposed community downstream of Pittsboro; in residents of Wilmington, North Carolina, PFNA and PFOS were associated with higher total and non-HDL cholesterol (Rosen et al., 2022). Conversely, our results for PFHxS conflict with some studies that have found negative associations between PFHxS and total and non-HDL cholesterol (Nelson et al., 2010), though these associations have been inconsistent over time (Dong et al., 2019). However, unlike other studies, we did not account for or exclude participants who were taking cholesterol-lowering medications, and our study population is much smaller in number.

3.6. Serum clinical chemistry

The associations between blood PFAS and various clinical chemistry health measures were evaluated. Health measures in the comprehensive metabolic panel (CMP) include biomarkers of liver and kidney function, such as ALT and AST. The full list of health measures is described in Table 4. PFAS exposure has previously been linked to altered liver function biomarkers (Gallo et al., 2012) and is associated with liver injury and disease (Costello et al., 2022; Ducatman and Fenton, 2022).

Spearman correlations for serum PFAS and serum clinical chemistry were calculated (Supplemental Fig. S12) with data stratified by timepoint (T1, $n = 48$; T2, $n = 44$; total, $n = 92$). Associations between serum PFAS and serum clinical chemistry were further evaluated using repeated-

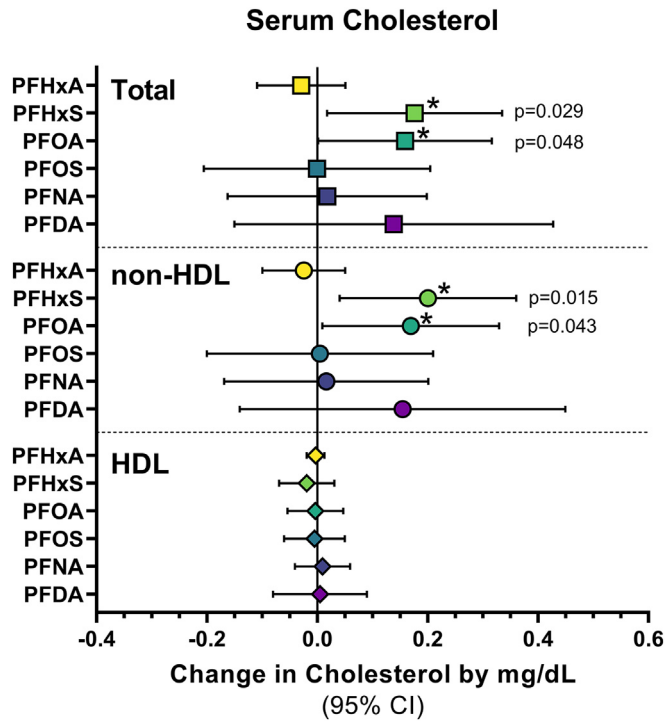

Fig. 2. Associations between serum PFAS and serum cholesterol. A 95% confidence interval (95% CI) is reported. Depicts change in cholesterol for each 1% increase in serum PFAS. Analyses were adjusted for age and sex. PFAS serum concentrations were $\log_{10}$-transformed before analysis. *$p < 0.05$, $n = 92$; repeated-measures, linear mixed models. Note that HDL cholesterol typically accounts for 20–30% of the total cholesterol.

Table 4
Health measures included in the serum comprehensive metabolic panel (CMP). Abbreviation, units, and broad functional category of the measure are listed. Values from each participant ($n = 49$) were averaged from both timepoints; arithmetic means and standard deviation (SD) for each endpoint are reported.

Function	Health measure	Abbreviation	Units	Mean ± SD
Glucose	Glucose	–	mg/dL	92.3 ± 17.2
Kidney tests	Blood urea nitrogen	BUN	mg/dL	16.4 ± 5.0
	Creatinine	Cre	mg/dL	1.0 ± 0.2
	Ratio of blood urea nitrogen to creatinine	BUN:Cre	unitless ratio	17.3 ± 4.9
	Estimated glomerular filtration rate	eGFR	mL / min / 1.73 m^2	76.5 ± 16.2
Electrolytes	Calcium	Ca	mg/dL	9.7 ± 0.4
	Chloride	Cl	mmol/L	101.1 ± 2.3
	Sodium	Na	mmol/L	138.2 ± 2.1
	Potassium	K	mmol/L	4.5 ± 0.4
	Carbon dioxide	CO2	mmol/L	22.7 ± 1.3
Liver tests	Alkaline phosphatase	ALP	IU/L	76.0 ± 18.3
	Aspartate aminotransferase	AST (or SGOT)	IU/L	21.3 ± 6.5
	Alanine aminotransferase	ALT (or SGPT)	IU/L	20.8 ± 9.7
	Bilirubin	–	mg/dL	0.4 ± 0.2
Protein	Protein, Total	–	g/dL	7.0 ± 0.3
	Albumin	–	g/dL	4.5 ± 0.2
	Globulin, Total	–	g/dL	2.5 ± 0.3
	Ratio of albumin to globulin	A:G Ratio	unitless ratio	1.9 ± 0.3

measures linear mixed models. Significant model results are summarized in Supplemental Table S8, and beta-coefficients and full model results are reported in Supplemental Table S7. Serum PFHxS, PFOA, PFNA, and PFDA were not statistically associated with any of the clinical chemistry measures. Serum PFHxA was statistically negatively associated with ALP, AST, carbon dioxide, chloride, and sodium, whereas serum PFOS was statistically positively associated with BUN:Cre ratio and glucose. Associations between all outcomes and age or sex (unadjusted) are reported in Supplemental Table S9.

Our limited sample size and relatively homogeneous cohort may have limited our ability to detect associations. However, given that most of our statistically significant associations were with PFHxA, it is worth highlighting this compound for future study in larger, more robust epidemiological studies. This is especially relevant since PFHxA was the dominant PFAS measured in drinking water in this study, and PFHxA is not frequently detected in the blood of the U.S. population (Calafat et al., 2006; 2022). While the exposure and health effects of PFHxA in humans have been infrequently studied, previous studies in animals have shown that chronic PFHxA exposure is associated with kidney effects such as mild renal degeneration, mild to severe papillary necrosis, and increased urine volume (ATSDR, 2021; Klaunig et al., 2015).

The National Academies of Sciences, Engineering, and Medicine (NASEM) issued a Guidance on PFAS Exposure, Testing, and Clinical Follow-Up in 2022 (NASEM, 2022). In their guidance, NASEM recommended actions based on the sum of serum concentrations for seven PFAS (i.e., blood Σ(7)PFAS = sum of PFOS, PFOA, PFHxS, PFNA, PFDA, PFUnDA, and MeFOSAA). While our testing in this study included only five of these seven analytes, these five PFAS were detected in the blood of every study participant (i.e., 100% detection frequency; Table 2). All of our participants ($n = 49$) had a blood Σ(6)PFAS over 2 ng/g (Supplemental Table S10). The median blood Σ(6)PFAS concentration was 25.3 ng/g; thus, over 50% of our participants would fall under NASEM's >20 ng/mL category for the highest level of medical monitoring.

3.7. Limitations

There are a few limitations to this work that should be noted. One limitation is the lack of information regarding historical drinking water consumption. From previous studies, we know Pittsboro's drinking water previously contained higher concentrations of PFAAs (Nakayama et al., 2007; Sun et al., 2016a), and in this study we observed higher blood levels of PFOA in older participants and in participants who had lived in their current home for 7 years or longer. Additional survey questions could have helped us better gauge historical drinking water exposure (e.g., how long have they lived in Pittsboro, what year did they regularly start filtering their water).

Additionally, the small sample size and homogeneity of this cohort limits some analyses, such as analysis of participants with eumenorrhea or participants without water filtration. The health measures data are limited as measurements were obtained from non-fasting participants; associations with measures that can be affected by fed state (e.g., triglycerides, glucose) may be impacted as our participants were not fasting. We also did not account for other health conditions or medications such as cholesterol-lowering medications. Our relatively small and homogeneous cohort also precluded further analysis of health measures by age, sex, or menstrual status.

4. Conclusions

Pittsboro residents in this study were found to have blood PFAS concentrations at significantly higher levels than the U.S. population and at levels similar to downstream communities that also have drinking water PFAS contamination. The Haw River has historically had considerable levels of PFOA and PFOS contamination, at least since 2006 (Nakayama et al., 2007; Sun et al., 2016a), and bluegill fish collected from the Haw River in 2007 had median PFOS levels of 30 ng/g fillet (Delinsky et al., 2009). This historical PFAS contamination may explain why the levels of serum PFAS are currently elevated in Pittsboro residents.

We also found some statistically significant associations between serum PFAS and health measures of cholesterol and liver biomarkers. Serum PFOA and PFHxS were both associated with increased total and non-HDL cholesterol, and serum PFOS and PFHxA were associated with differences in clinical chemistry. It is important to note that the associations observed here between PFAS exposure and total and non-HDL cholesterol are similar to recent observations in the downstream community of Wilmington, NC that included over 300 participants (Rosen et al., 2022). This study suggests that more research may be needed to understand health impacts in communities that have elevated exposures to PFAS in their drinking water.

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CRediT authorship contribution statement

SMH: Study design; Sample collection; Investigation; Data analysis; Visualization; Writing – original draft; Writing – review & editing; Writing – revising; **SZ:** Investigation; **GHT:** Investigation; **KH:** Conceptualization; Study design; Sample collection; Data analysis; Writing – review & editing; **DNC:** Data interpretation; **JAH:** Study design; Data interpretation; **HMS:** Conceptualization; Study design; Sample collection; Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

De-identified data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.165091.

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